CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EE) Examination

November 2012

EE 203 ELECTRICAL MEASUREMENT AND MEASURING INSTRUMENTS

Date: 07.11.2012, Wednesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain briefly Moving Iron type Synchroscope. [04]

(b) Write a short note on Phase Sequence Indicators. [03]

Q - 2 (a) State and explain different causes of errors in Current Transformers. [04]

(b) Explain effect on characteristics of Current Transformer with respect to change in following parameters: [06]

- Change of Primary winding Current
- Effect of P.F. of Secondary Winding Burden on errors
- Effect of Change of Frequency

(c) Explain briefly any one type of loop test used for location of ground and short circuit faults in cables. [4]

OR

Q - 2 (a) Explain with a neat diagram Absolute null method for testing of Potential Transformers. [05]

(b) In a test for a fault to earth by Murray loop test, the faulty cable has a length of 5.2 Km. The faulty cable is looped with a sound cable of the same length and cross-section. The resistances of ratio arms are 100 Ω and 41.2 Ω at balance. Calculate the distance of the fault from the test end. If the decade resistance boxes forming the ratio arms have limits of error of ± 0.5 % (standard deviation) of the dial reading, what is the limit of error in the above calculated result? [04]

(c) Discuss briefly construction of Current Transformers [05]

Q - 3 (a) Explain in detail construction of Permanent Magnet Moving Coil (PMMC) type instruments. [06]

(b) Explain different types of errors that occur in Moving Iron type of Instrument. [04]

(c) State and explain advantages and disadvantages of rectifier type Instruments. [04]

OR

Q - 3 (a) Derive general torque equation for PMMC type Instrument. [06]

(b) Explain in detail advantages and disadvantages of Moving Iron type Instruments. [04]

(c) Derive force and torque equations of Electrostatic Instruments. [04]

SECTION – II

Q - 4 (a) Explain briefly following terms related to measuring instruments: [04]

- Sensitivity
- Accuracy

(b) State and explain briefly different types of errors in measurements. [03]

Q - 5 (a) Explain operation of basic slide wire potentiometer. Also explain procedure for its standardisation. [07]

(b) What are the difficulties faced in measurements of high order resistance. Also explain used of guard circuit in the measurement of high order resistance. [07]

OR

Q - 5 (a) Explain in detail Duo-range type potentiometer. [07]

(b) Explain Kelvin's double bridge for measurement of low order resistance. [07]

Q - 6 (a) Explain with a neat sketch and phasor diagram Maxwell's Inductance bridge. [05]

(b) Write a short note on Anderson's bridge. [05]

(c) Write a short note on Measurement of Flux Density in ring specimen. [04]

OR

Q - 6 (a) Explain with a neat sketch and phasor diagram De Sauty's bridge. [05]

(b) An Owen's bridge shown in Fig. 1 is used to measure the properties of a sample of sheet steel at 2 KHz. At balance, arm ab is test specimen; arm bc is $R_3 = 100\ \Omega$; arm cd is $C_4 = 0.1\ \mu F$ and arm da is $R_2 = 834\ \Omega$ in series with $C_2 = 0.124\ \mu F$. Derive balance conditions and calculate the effective impedance of the specimen under test conditions. [05]

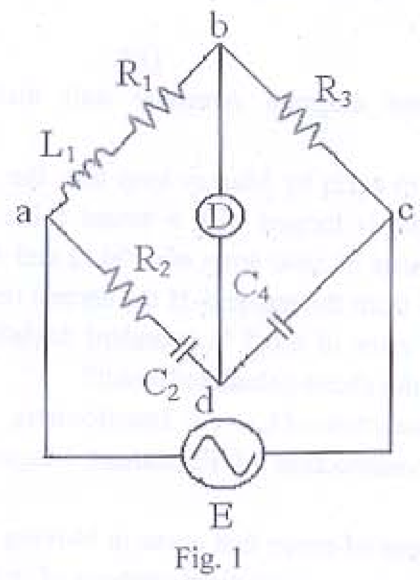

Fig. 1

(c) Write a short note on Magnetic Potentiometer. [04]

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